

Data Preprocessing for Machine Learning

Handling Missing Values

- $< 5\%$ ---> Drop
- $5\% - 40\%$ ---> Fill
- $> 40\%$ ---> Drop Column

Handling Categorical Data

- Ordinal ---> Mapping or OrdinalEncoder
- Nominal ---> - OneHotEncoder
 - > 7 categories (BinaryEncoder)

	occupation_0	occupation_1	occupation_2	occupation_3
0	0	0	0	1
1	0	0	1	0
2	0	0	1	1
3	0	1	0	0
4	0	0	1	0

2 ³	2 ²	2 ¹	2 ⁰
2	2	2	2
0	0	0	1
0	0	1	0 ←
0	0	1	1
0	1	0	0
0	0	1	0 ←
1	0	0	1

Gender

F
M

	F
	M
	M
	F

OITE

	F	M
	1	0
	0	1
	0	1
	1	0

Age, gender, salary, KPI, promoted



ML
Supervised
Target Column



Continuous
Regression

Categorical
Classification

Feature Scaling

Apply on Numerical Data

MinMax Scaler

$$\frac{(x_i - \min)}{\max - \min}$$

Standard Scaler

$$\frac{(x_i - \mu)}{\sigma}$$

Robust Scaler

$$\frac{(x_i - \text{median})}{Q_3 - Q_1}$$